

BENNETT JOSEPH DANIEL

Richardson, TX | bennett.daniel@utdallas.edu | [Linkedin](#) | [Portfolio](#)

SKILLS

Technical Skills: MATLAB, Simulink, Simscape, OptimumLap, OptimumDynamics, CarMaker, ANSYS, SolidWorks

Automotive: Vehicle System Design, ADAS (path planning), CAN, SIL & HIL Testing.

Control Systems: Observer Design, Differential Flatness, PID Control, Model Predictive Control, Optimal Control

Programming and Analysis Tools: Python, ROS2, C++, Linux, TensorFlow, Scikit-Learn, Pandas, Matplotlib, git, Jira

Model Training and Deployment: YOLO, CNN, Object Detection & Classification, Reinforcement Learning, Deep Learning

WORK EXPERIENCE

NOVA (UT Dallas Autonomous Driving Group)

Richardson, TX

Master's Thesis Student

Aug 2023 – Present

- Conducted comparative **analysis** of five cost-based path planning algorithms within a **ROS2** architecture
- Implemented and validated **path planners** using SIL testing in CARLA with standardized self-driving scenarios
- Evaluated performance to enhance Navigator, UTD's open-source autonomous driving stack

COMET SOLAR RACING

Richardson, TX

Vehicle Dynamics Lead Engineer

Jan 2024 – Present

- Developed ESC using **MATLAB/Simulink** and **Simscape**, integrating vehicle and brake dynamics in IPG CarMaker
- Built a custom lap simulator to analyze and tune solar car performance under dynamic conditions
- Reduced brake system unsprung mass by 4 kg; optimized suspension design using SolidWorks and FEA in ANSYS
- Designed and assembled suspension and steering systems with GD&T-based quality control for precise handling

AUTOMOTIVE RESEARCH CELL

Chennai, India

Research Assistant

Sep 2022 – May 2023

- Assisted Dr. V. Ganesh in **automotive safety research**, focusing on vehicle accident analysis
- Developed and **published** a novel method for classifying accident-related injuries
- Designed a SolidWorks-based **hardware device** to reduce emergency response time in vehicles

ARTENAL

Remote

Deep Learning Intern

Aug 2022 – Nov 2022

- Annotated 1,000+ images to train computer vision algorithms for waste segregation.
- Trained **YOLO** and **SSD** models for accurate **image classification** with 95% accuracy.
- Deployed a computer vision model to streamline operations.

IIIT HYDERABAD (Robotics Research Center)

Remote

Summer Research Intern (SRISHTI-22)

May 2022 – Jun 2022

- Acquired foundational knowledge of modeling for robotics.
- Coded and validated basic manipulator movements using the ROS architecture.

PUBLICATION

SMART ACCIDENT FATALITY REDUCTION (SAFR) SYSTEM

Joseph, D.B. et al. (2023). Smart Accident Fatality Reduction (SAFR) System. In: Kumar, S., Sharma, H., Balachandran, K., Kim, J.H., Bansal, J.C. (eds) Third Congress on Intelligent Systems. CIS 2022. Lecture Notes in Networks and Systems, vol 613. Springer, Singapore. https://doi.org/10.1007/978-981-19-9379-4_38

EDUCATION

University of Texas at Dallas

Richardson, TX

Master of Science in Mechanical Engineering

Expected May 2025

Anna University

Chennai, India

Bachelor of Engineering in Automobile Engineering

2019 - 2023